 NAME ____________ DATE ____________

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1 G protein coupled receptors

COMP

You should be able to: Trace a G protein coupled receptor pathway from ligand binding through the G protein and amplifier enzyme to the second messenger and the cellular response.

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2 Second messengers

COMP

You should be able to: Identify cyclic AMP, IP3, diacylglycerol, and calcium as second messengers and state the enzyme that generates each and one target it activates.

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3

COMP

Catalytic receptors and intracellular receptors

You should be able to: Compare a receptor enzyme such as the insulin receptor with an intracellular steroid receptor by mechanism and by time course of the response.

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4

COMP

Signal termination

You should be able to: Name three mechanisms that end a chemical signal and explain why a signal that cannot be terminated produces pathology.

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5

COMP

Dose response relationships

You should be able to: Plot a dose response curve, identify threshold, maximal response, and EC50, and compare a full agonist with a partial agonist on the same axes.

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6

COMP

Hormone classes

You should be able to: Classify hormones as peptide, steroid, or amine and compare the three by synthesis, storage, and receptor location.

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7

COMP

Hormone transport and half life

You should be able to: Explain how protein binding in plasma affects the free hormone fraction and the half life and predict which hormone class circulates bound.

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8

COMP

Hormone receptors and target response

You should be able to: Explain why one hormone can produce different responses in different tissues and relate receptor number to target cell sensitivity.

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9 COMP — Hormone interactions

You should be able to: Define synergism, permissiveness, and antagonism and identify each in a described hormone pair.

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10 COMP — Control of hormone release

You should be able to: Classify a hormone stimulus as humoral, neural, or hormonal and trace an example of each.

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11 COMP — Hypothalamic pituitary axes

You should be able to: Trace a three tier axis from hypothalamic releasing hormone through the anterior pituitary trophic hormone to the peripheral gland and place the long and short loop negative feedback.

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12 COMP — Primary and secondary endocrine disorders

You should be able to: Use hormone levels at two tiers of an axis to classify a disorder as primary or secondary and as hypersecretion or hyposecretion.

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13
COMP

Posterior pituitary hormones

You should be able to: Explain why the posterior pituitary is neural tissue and state the stimulus and target action of antidiuretic hormone and oxytocin.

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14
COMP

Hormone assay

You should be able to: Run or simulate an immunoassay for a hormone and use the standard curve to determine an unknown concentration.

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